AI now outperforms humans on many tasks, yet working with it rarely makes people more effective, more productive, or more satisfied. Human-AI collaboration breaks down on three fronts: at the person, who treats the model as if it thinks and in this way misplaces trust in its outputs; at the model, which errs in ways no person would, so its mistakes slip past the people checking it; and at the interaction between them, where small design choices decide whether AI helps or gets in the way. Worse, we lack a clear methodology for studying any of these failures, so the field improvises with common-sense intuitions.

My research confronts all three fronts. I build methods to measure human-AI interaction, use them to learn what people actually need, and engineer systems that meet those needs. Along the way, I have shown that explanations fail as safeguards against discrimination and that interpretability can increase overreliance in decision-making settings. I have also built methods use reinforcement-learning policies to measurably improve human planning.

Here, I outline two projects that extend this agenda to generative AI. In the first, I build methods to measure how people interact with generative systems to learn which interface designs best serve users. In the second project, I engineer a system that learns from user feedback to personalize AI outputs more efficiently. Together, these projects bring generative AI closer to actual user needs.

To date, my work has fostered international collaborations across machine learning, cognitive science, philosophy, and public policy, with publications at top venues including Machine Learning, Behavior Research Methods, Cognitive Science, and FAccT.

Research Areas: Human-Centered AI, AI for Ethics & Fairness, Interactive ML & Agents

Proposed Research

Area: Human-Centered AI

Specializing LLMs Without Fine-Turning

The next wave of AI value comes from deploying language models as specialized tools - medical advisors, coding assistants, legal analysts. Our current technique iterates between generating model responses, scoring them against rubrics, i.e., weighted criteria such as "safe dosing recommendation" or "step-by-step calculation" for drug dosing advice, and updating the model based on those scores. The key challenge in this technique is creating rubrics. In practice, most rubrics are designed by domain experts who manually assign weights and criteria based on intuition and heuristics. This process requires thousands hours of work per deployment, provides no systematic way to handle disagreement between experts, and offers no guarantees that the resulting criteria actually optimize for desired outcomes.

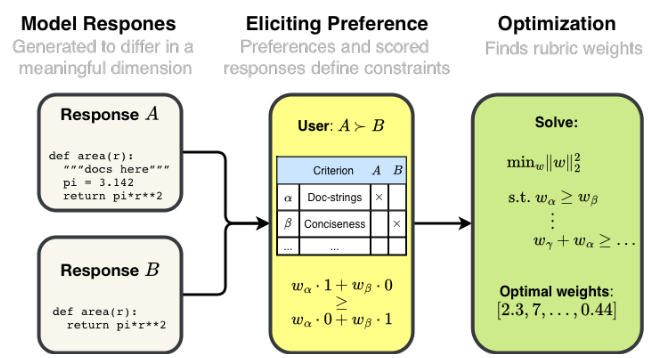

Fig. 1: Converting user preferences into optimized rubric weights.

In this project, I will standardize rubric design by creating a scalable, automated method for learning optimal rubric weights from user preferences (see Fig. 1). My approach will associate each rubric criterion $c : R \to \{0,1\}$, where $c(r)$ indicates whether response r satisfies criterion c, with a unit vector v_c. I will gather user preferences on pairs of model responses: $r \succeq r'$. Each model response r can be then represented as a binary vector in the criteria space, i.e., as $V_r = \sum_{\{c:\, c(r)=1\}} v_c$. This representation allows translating user preferences into linear inequalities: if $r_c \succeq r'_c$ then $w^\top V_{r_c} \geq w^\top V_{r'_c}$ where w represents the rubric's weights. Satisfying these inequalities means that the rubric scores responses that humans prefer higher than those they do not prefer. We can use these inequalities to solve an optimization problem and find the weights that satisfy them. This framework extends naturally to multi-stakeholder alignment. When preference conflicts arise, we could use selective aggregation and establish partial orderings only when preferences have enough support in the data.

My hypothesis in this project is that defining rubrics via optimization will produce evaluations more aligned with expert intent than rubrics defined heuristically by experts. Because optimization offers a principled, straightforward approach to rubric creation, this research can significantly speed up rubric generation and provide a scalable solution for creating specialized, personalized models.

Designing for Engagement in Generative AI Models

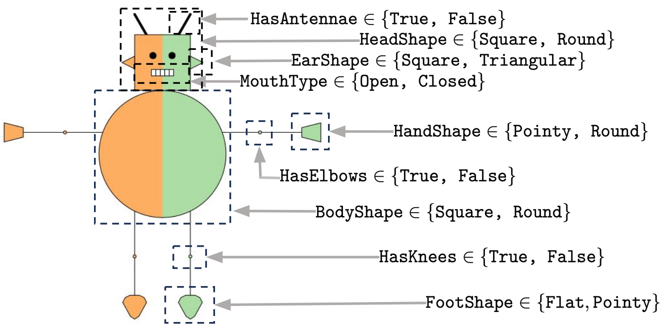

Fig. 2: Example robot with features users could specify when generating robots through our proposed generative AI platform. This controlled generation task enables systematic testing of interface design choices with clear success criteria and reproducible experimental conditions.

While generative AI systems have become increasingly powerful, we lack a systematic understanding of how users actually interact with these tools in practice. Current generative AI interfaces offer multiple input modalities, including text descriptions, concept selection, style choices, and reference images. However, we don't know which combinations are most effective, where users struggle, or what causes abandonment. Without empirical evidence of user behavior patterns, interface designers often rely on intuition rather than data, which limits their ability to optimize the user experience.

In this project, drawing on my experience creating experimental platforms for studying interpretability [SGU24] and discrimination detection [SDU25], I will develop a platform to test how users engage with different generative AI interface designs. To design the platform, I will extend my existing robot classification environment [SGU24] by enabling users to request images of robots with specific characteristics. This controlled environment provides clear success metrics (achieving specific robot features) and eliminates confounding variables that complicate open-ended creative tasks (user intent, domain expertise, etc.). Through a series of studies, I will investigate which design choices optimize task completion speed, output quality, and sustained engagement. Initial studies will focus on two key dimensions: (1) concept specification methods (natural language vs. structured controls vs. example-based) and their effects on task completion and output quality, and (2) iteration workflows (real-time editing vs. generate-then-refine vs. conversational steering) and their effects on engagement and abandonment rates. The platform will capture detailed interaction logs, including the time spent, revision sequences, and points of abandonment. My hypotheses are that: (1) structured controls will enable faster convergence for specific targets while natural language will better support exploratory tasks, and (2) real-time editing will reduce abandonment by providing immediate feedback. This research aims to establish evidence-based design principles for generative AI interfaces, transitioning the field to data-driven design decisions.

Past Research — Areas: AI for Ethics & Fairness, Interactive ML & Agents

Effect of Interpretability on Human-AI Collaboration [SGU24]

A core assumption in human-AI interaction is that interpretable models lead to better human decision-making. The rationale is that knowing how the model maps inputs to outputs helps identify model errors, biases, or spurious correlations, and helps users recognize when they have side information. This assumption is problematic to validate because results may not generalize across use cases (e.g., clinical vs. judicial decision-making), may be influenced by people's prior beliefs, and may depend on comparisons to optimal decisions under available information.

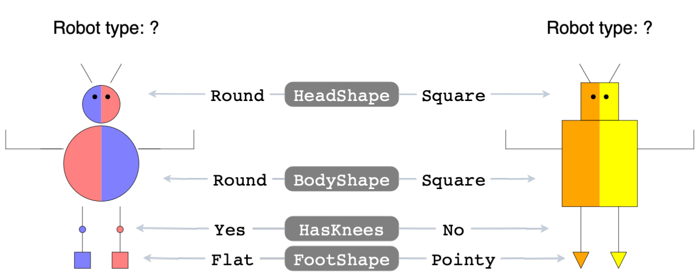

Fig. 3: In our platform, a model predicts robot type based on four binary characteristics, ($\mathbb{1}[\texttt{HeadShape} = \texttt{Round}]$, $\mathbb{1}[\texttt{Bodyhape} = \texttt{Round}]$, $\mathbb{1}[\texttt{Knees} = \texttt{Yes}]$, $\mathbb{1}[\texttt{FootShape} = \texttt{Pointy}]$). We investigate whether, once the model is interpretable and explicitly shows how robot features predict robot type, people make more accurate predictions themselves.

Linear Classifier $\hat{y} = 1 \text{ if } 1.74x_1 + 1.10x_2 \geq 2.84$

Scoring System $\hat{y} = 1 \text{ if } x_1 + x_2 \geq 2$

Boolean Rule $\hat{y} = x_1 \text{ AND } x_2$

Fig. 4: In our platform, we represent models in interpretable formats like a linear function (top) or a scoring system (a linear function with integer coefficients that can be treated as a weighted checklist; middle). We found that experts' reliance on the models varies by format, leading to different degrees of overreliance.

In this project, I co-ideated a fully controllable setup (robot-type prediction, as shown in Fig. 3), computed all linearly separable models in the environment, and designed quantitative metrics to compare models and measure the effect of interpretable models on decision-making (models depicted in Fig. 4). In my setup, I can set the model's accuracy, format, and its relationship to the model that describes the user's beliefs, among other parameters, and define the default reliance on the model's predictions. I used this setup to show that humans overrely on interpretable models, regardless of accuracy, and that reliance changes with the chosen model format (as shown in Fig. 4) [SGU24]. We are currently conducting additional studies and plan to submit our work to an HCI venue (CHI) and a general science journal (PNAS).

Improving Human Decision-Making Through Automatically Discovered Decision Aids [SBL21; BSOL22]

A fundamental challenge in human-AI interaction is that users often cannot harness the benefits of AI systems because they cannot understand them. Reinforcement learning (RL) is one area where insights from AI could help both experts (e.g., in sepsis treatment) as well as the general public (to learn clever heuristics for decision-making). The primary issue is that RL generates opaque, stochastic black-box policies that offer no insight into the underlying decision-making logic.

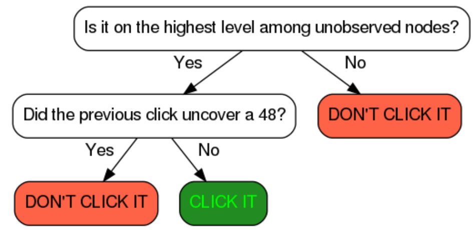

Fig. 5: Automatically discovered heuristic in a tree-graph environment where nodes encode outcomes of actions (the higher, the more long-term). The strategy assumes inspecting all long-term outcomes until it finds the best one.

To clarify the end product of RL, I developed a Bayesian imitation learning algorithm that outputs interpretable descriptions of policies as flowcharts (as in Fig. 5) [SBL21]. My approach combines advances in imitation learning and program induction with a novel clustering method to identify subsets of demonstrations that simple, high-performing decision rules can accurately describe. The key innovation is that my algorithm can handle cases where traditional methods fail, i.e., when the created domain-specific language is insufficient to describe the whole set of demonstrations or when policies exhibit idiosyncratic behaviors that cannot be captured by available language. Through extensive behavioral experiments, I demonstrated that humans can successfully understand and apply the flowcharts generated by my method, significantly improving their strategies across different sequential decision problems. Most importantly, my approach outperformed conventional training methods that rely on performance feedback, showing that AI-discovered strategies can transfer to human decision-makers with appropriate interpretability tools.

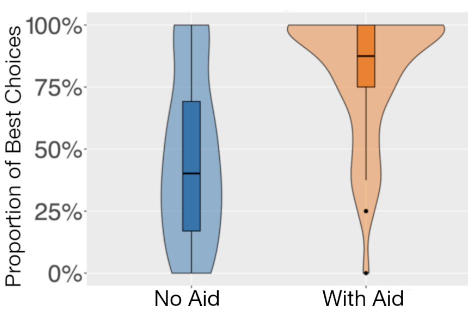

Fig. 6: Proportion of choices for the mortgage with the lowest interest rate when participants can use the decision aid versus when they cannot.

I then extended my work to create practical interventions that help humans apply AI-discovered heuristics in naturalistic settings [BSOL22]. I focused on promoting far-sightedness – countering human tendency to prefer immediate gains over long-term rewards – in two tasks: planning a road trip and choosing a mortgage. My key finding was that the intervention format matters critically, as procedural instructions (step-by-step guides) led to significantly better outcomes than static flowcharts. This finding motivated my primary technical contribution — an algorithm that transforms flowcharts into procedural instructions using linear temporal logic. My empirical contribution is that decision aids generated with this algorithm promote better decisions (see ??). This work shows that while humans can benefit from AI-discovered strategies, success depends on translating them into formats that align with how people naturally think and process information.

Evaluating Explanations for Algorithmic Discrimination [SDU25]

One challenge in human-AI collaboration is that technological progress outpaces empirical validation. This is especially important for research in explainable AI as regulatory frameworks increasingly mandate model explanations as safeguards against algorithmic discrimination (see the example in Fig. 7), assuming humans can reliably use these explanations to detect unfair predictions. However, this assumption had never been rigorously tested under controlled conditions, because discrimination detection is inherently a probabilistic task without a clear ground truth. Detection is also prone to human errors tied to flawed prior knowledge.

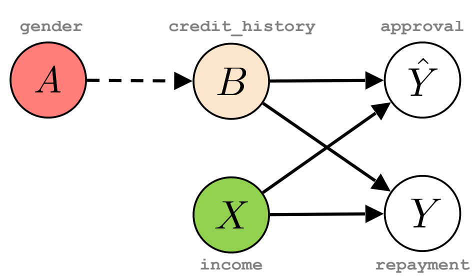

Fig. 7: Causal diagram for discrimination detection. For example, in loan approval predictions ($\hat{Y}$), the model uses an individual's income (X) and credit history (B) as inputs. Gender (A) could affect credit history due to differences in credit scores or the intensity of credit usage found between men and women.

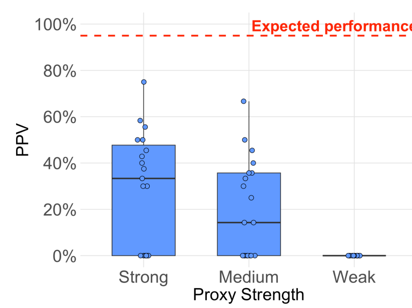

Fig. 8: Reliability of discrimination detection with explanations. We expect people to spot $\geq 95\%$ of discrimination (red line). Instead, people retrieve only 50% of all unfair predictions, irrespective of how strong the proxy for the protected attribute is.

In this work, I developed synthetic controlled conditions to test these assumptions, ultimately conveying the limitations of explanations for assessing fairness [SDU25]. I created a framework for judging the fairness of predictions at an instance level through counterfactual fairness theory, where a prediction is (δ-)counterfactually fair if changing the protected attribute can change the probability of obtaining the same prediction (by at most δ). I created a synthetic environment where I could control failure modes of discrimination detection unrelated to explanations: whether people know the proxies, their strength, the protected class for predictions they study, whether they know how to use explanations to support claims, and whether explanations hide relevant information.

I utilized my framework to empirically demonstrate that explanations are unreliable for assessing discrimination. Even with perfect knowledge of proxy strength, protected-class participants retrieve between 40% and 70% of the discriminatory predictions (see Fig. 8) and systematically label 30% of all fair predictions as discriminatory, revealing their incorrect assumptions about proxy identity. This work provides evidence that current regulatory approaches that rely solely on explanations are insufficient, underscoring the broader need to empirically validate assumptions about human-AI collaboration. To date, it has drawn interest from governmental agencies in the US, the Netherlands, and Switzerland.

Evaluating Concept Bottleneck Models [SCKSU25]

Concept bottleneck models (CBMs) are deep learning models that detect human-interpretable concepts in the input data and use them to make final predictions (see Fig. 9). This bottleneck allows humans to inspect the detected concepts and correct them to improve model performance. The greatest barrier to adopting these systems in practice is costly concept annotation. This investment would be justifiable if practitioners could thoroughly evaluate whether CBMs work for their specific use case. Yet, because reliable benchmarks with controlled variation are lacking, practitioners cannot make informed deployment decisions. In this project, we introduced synthetic benchmarks for two critical applications for CBMS: automating human work (where we used validation of Sudoku boards) and supporting human decision-making (where we used robot classification task from [SGU24]).

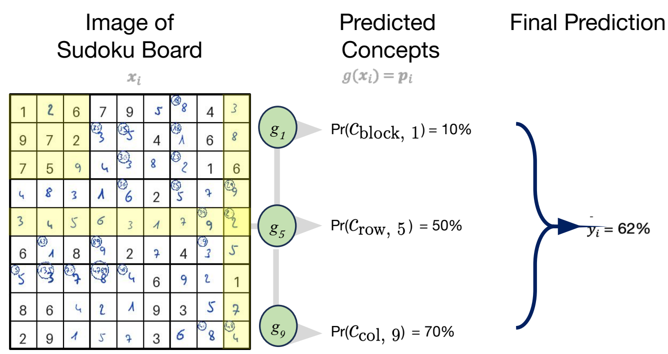

Fig. 9: Concept bottleneck models detect high-level features from the data, and then aggregate them to obtain the final prediction.

Our experiments revealed that while CBMs can outperform standard neural networks, they are highly sensitive to realistic deployment challenges. Results showed that overly granular concept sets reduced accuracy below the standard network performance (from 97.7% to 87.1% accuracy), concept annotation noise eliminated intervention benefits entirely (reducing gains from +16.3% to 0% accuracy), and expert constraints intended to improve model alignment actually degraded intervention effectiveness (reducing accuracy gains by over 10%). This work provides practitioners with the first systematic framework for evaluating when investments in CBMs are justified, illustrating how we can improve human-AI collaboration by improving evaluation of existing methods.

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